SUPPLY PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

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Warning
- **Fuel is highly flammable and dangerous. Fuel line spills and leakage can cause serious injury or death, and damage to equipment. If fuel comes into contact with the skin and eyes it will cause irritation. To prevent this, always perform the "Fuel Line Safety Procedure" referring to the "BEFORE REPAIR PROCEDURE".**
- **A person charged with static electricity could cause a fire or explosion, resulting in death or serious injury. Before servicing the fuel system, make sure to discharge static electricity by touching a vehicle.**

Caution
- **If parts are exposed to the fuel, their function may be decreased or a fire may occur. Spread a rag to absorb any leaked fuel.**
- **Do not let foreign matter get into the injection pipe to prevent clogging. Always tighten the injection pipe to the specified torque.**
- **To prevent fuel leakage, when removing/installing the supply pump, always follow the procedure in the Workshop Manual and perform the procedure.**

Note
- If the supply pump is replaced, perform the "Operation After Replacing Supply Pump".

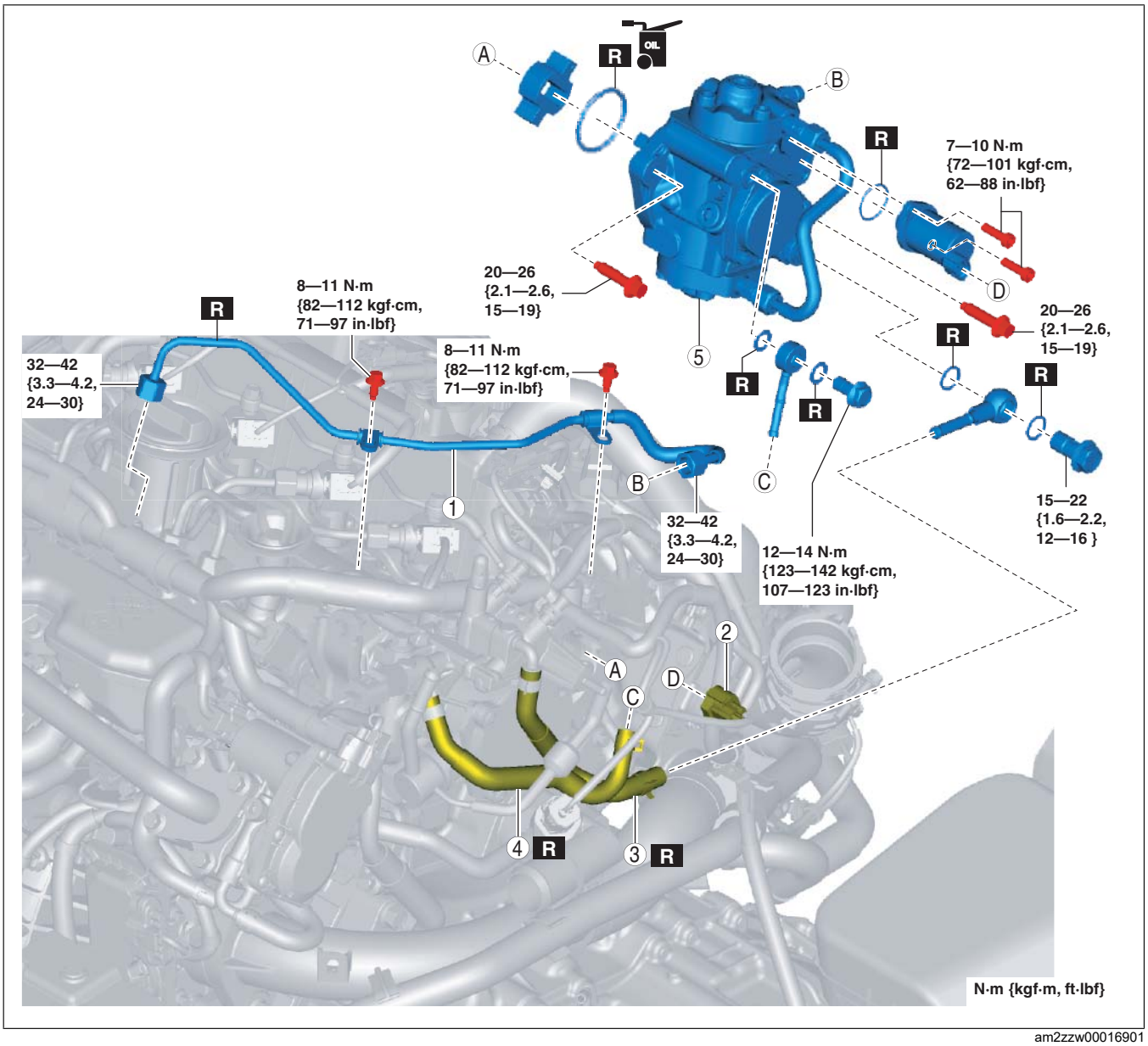
Operation After Replacing Supply Pump

1. If the supply pump is replaced, perform the following procedure.

STEP	ACTION	PAGE/CONDITION
1	Perform supply pump data reset procedure.	(See SUPPLY PUMP DATA RESET [SKYACTIV-D 1.5].)
2	Switch the ignition off.	—
3	Wait for 20 s or more.	—
4	Perform KOEO self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)
5	Maintain the idle status for 30 s with the following condition met. • ECT: 60—100 °C {140—212 °F}	—
6	Perform KOER self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)

Supply Pump Removal/Installation

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Complete the "BEFORE SERVICE PRECAUTION". (See BEFORE SERVICE PRECAUTION [SKYACTIV-D 1.5].)
4. Remove the following parts as a single unit: (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - Fresh-air duct
   - Air cleaner
   - Air hose
   - Air inlet pipe
5. Remove the battery and the battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Set the DC-DC converter (i-ELOOP) aside so that it does not interfere with the servicing. (With i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
7. Remove in the order shown in the figure.
8. Install in the reverse order of removal.
9. Complete the "AFTER SERVICE PRECAUTION". (See AFTER SERVICE PRECAUTION [SKYACTIV-D 1.5].)

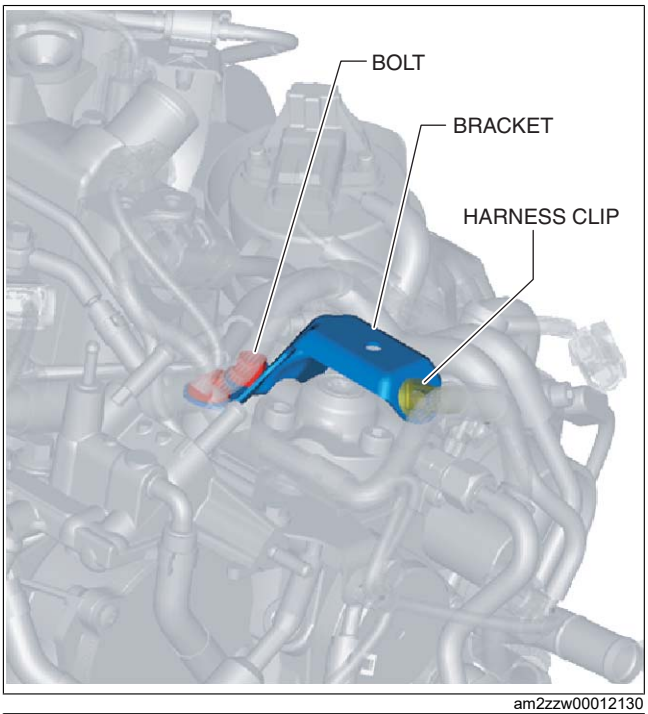


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1	Injection pipe (supply pump side) (See INJECTION PIPE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See Injection pipe (supply pump side) installation note.)
2	Suction control valve connector
3	Fuel main hose (See Fuel main hose installation note.)
4	Fuel return hose (See Fuel return hose installation note.)
5	Supply pump (See Supply pump removal note.) (See Supply pump installation note.) (See Operation After Replacing Supply Pump.)

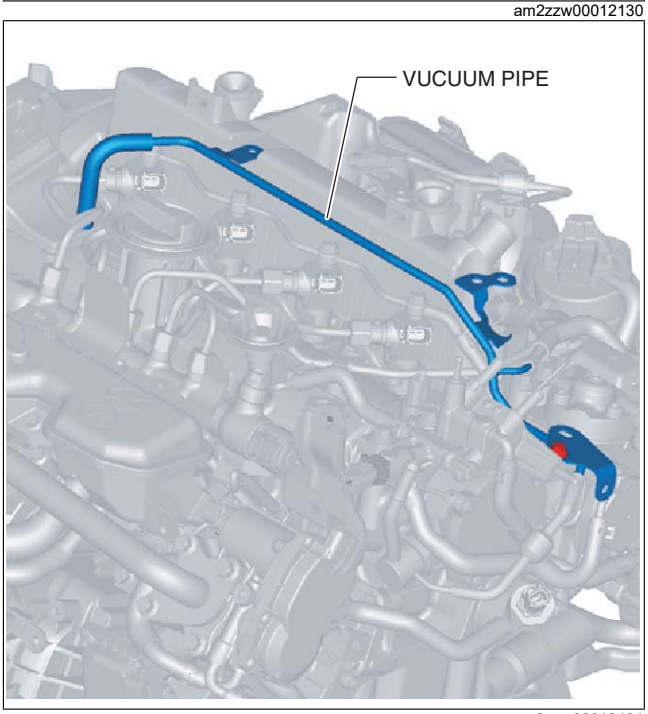
Supply pump removal note

1. Remove the wiring harness clip and bolt shown in the figure and set the bracket aside to a location which does not interfere with the servicing.
2. Disconnect the ECT sensor connector. (See ENGINE COOLANT TEMPERATURE (ECT) SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



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3. Remove the vacuum pipe.
4. Remove the supply pump.

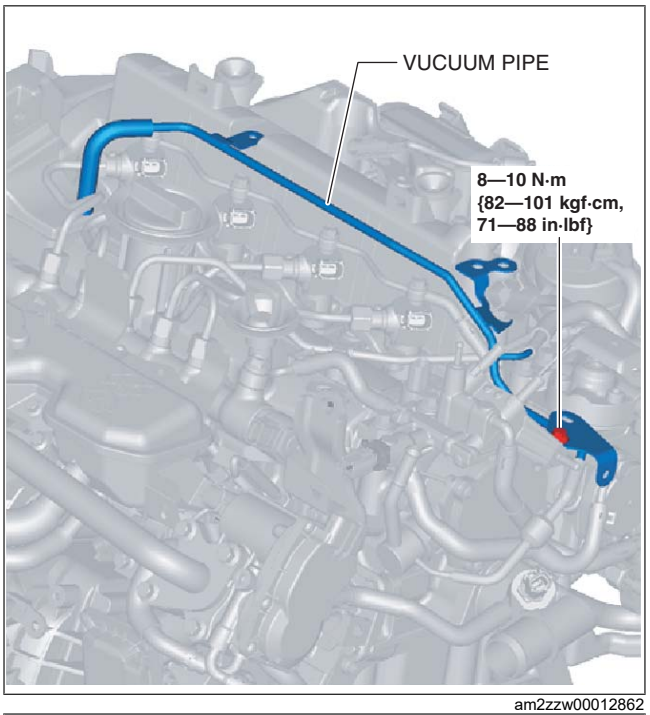


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Supply pump installation note

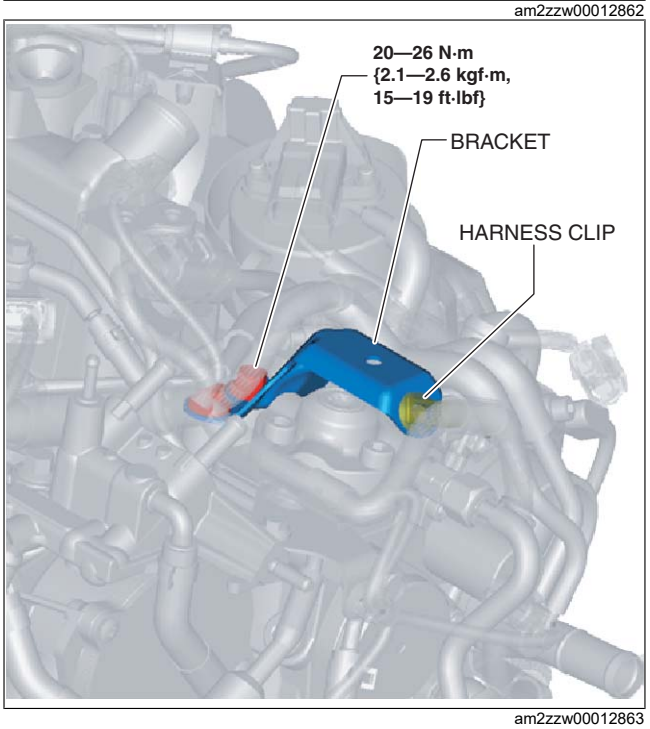
1. Install the supply pump.

2. Install the vacuum pipe.
3. Connect the ECT sensor connector. (See ENGINE COOLANT TEMPERATURE (ECT) SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



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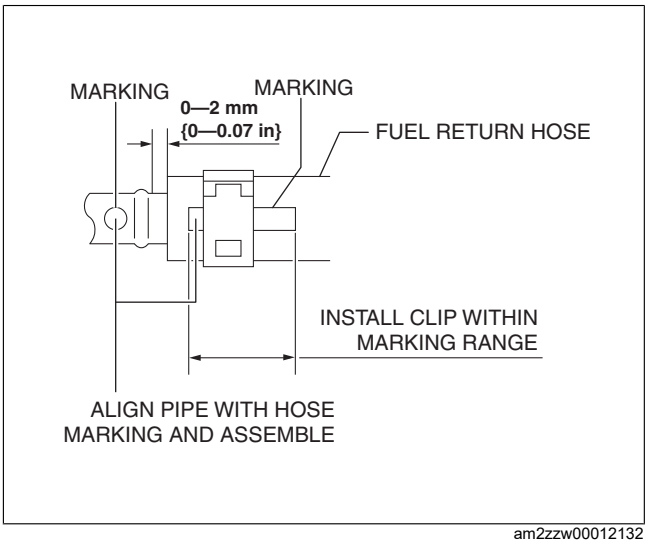
4. Install the bracket as shown in the figure.



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Fuel return hose installation note

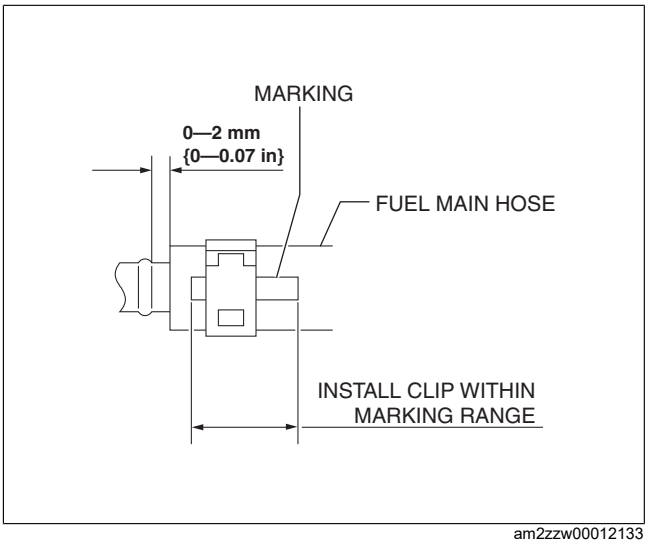
1. Install fuel return hose as shown in the figure.



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Fuel main hose installation note

1. Install fuel main hose as shown in the figure.



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Injection pipe (supply pump side) installation note

1. Temporarily tighten the injection pipe (supply pump side).
2. Temporarily tighten the injection pipe (supply pump side) bracket.
3. Tighten the injection pipe (supply pump side).

   Tightening torque
   32—42 N·m {3.3—4.2kgf·m, 24—30 ft·lbf}

4. Tighten the injection pipe (supply pump side) bracket.

   Tightening torque
   8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}